CSCE 4613
Artificial Intelligence

Review

Probabilistic Reasoning: Bayesian Networks

Spring 2026

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Announcements

- HW4 Grading – Released
- Midterm Grading - Released
- HW5 Submission Due: Apr 23
- Final Exam Review: April 30 (Last Day of Classes)
- **Final Exam Time: May 7, 2026**

Where We Are?

1. Introduction to AI (1 Week)
2. AI Programming Reviews (Python & Google Colab) (1 Week)
3. Search & Heuristics (2 Weeks)
4. Satisfiability (1 Week)
5. Knowledge representation (2 Weeks)
6. Probabilistic reasoning (1 Weeks)
7. Deterministic/symbolic reasoning (1 Weeks)
8. Sequential Decision Making (1 Weeks)
9. Neural Networks (1 Weeks)
10. Deep Learning Basics (2 Weeks)
11. A.I Applications (1 Week)

Recap

- Probability Models
  - Joint probability distribution of random variables

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- Probabilistic Inference
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- Chain Rule, Independence, Bayes' Rule
- Full joint distribution is hard to estimate and too big to represent explicitly

Outline

- Bayesian Networks

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- Independence in Bayes' Net

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- Applications of Bayes' Net

Bayesian Network

- Bayesian Network (Bayes' net) Overview

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    - A directed acyclic graph
    - Each node corresponds to a random variable
    - Each direct edge $\langle X_i, X_j \rangle$ represent X_i is a **parent** of X_j
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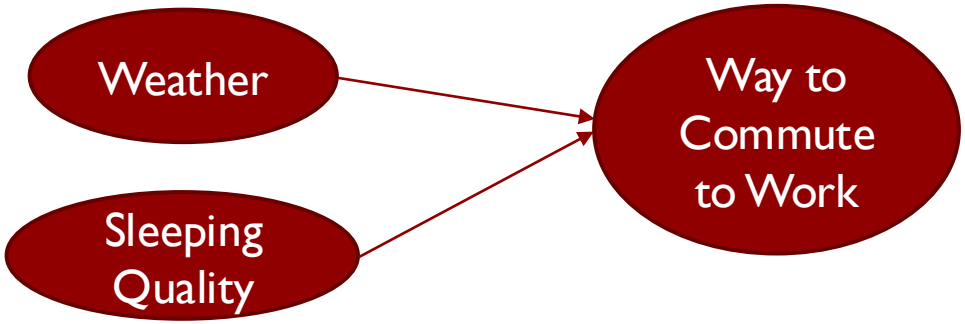
(≈ "directly influences")



Example: Alarm

- I'm at work, both of my neighbors John and Mary call to say my alarm for burglary is ringing. Sometimes it's set off by minor earthquakes. Is there a burglary?

- How do we model this scenario? How can we represent our knowledge in such a domain with uncertainty?

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T	T	T	T	F	$5.13e-7$
T	T	T	F	T	$1.33e-7$
…	…	…	…	…	…

- Task: Compute $P(Burglary = true | JohnCalls = true, MaryCalls = true)$

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- Recall Independence
  - $\mathbf{P}(X_1, X_2, \dots, X_n) = \mathbf{P}(X_1)\mathbf{P}(X_2) \dots \mathbf{P}(X_n)$

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- Are all the random variables independent in this example?

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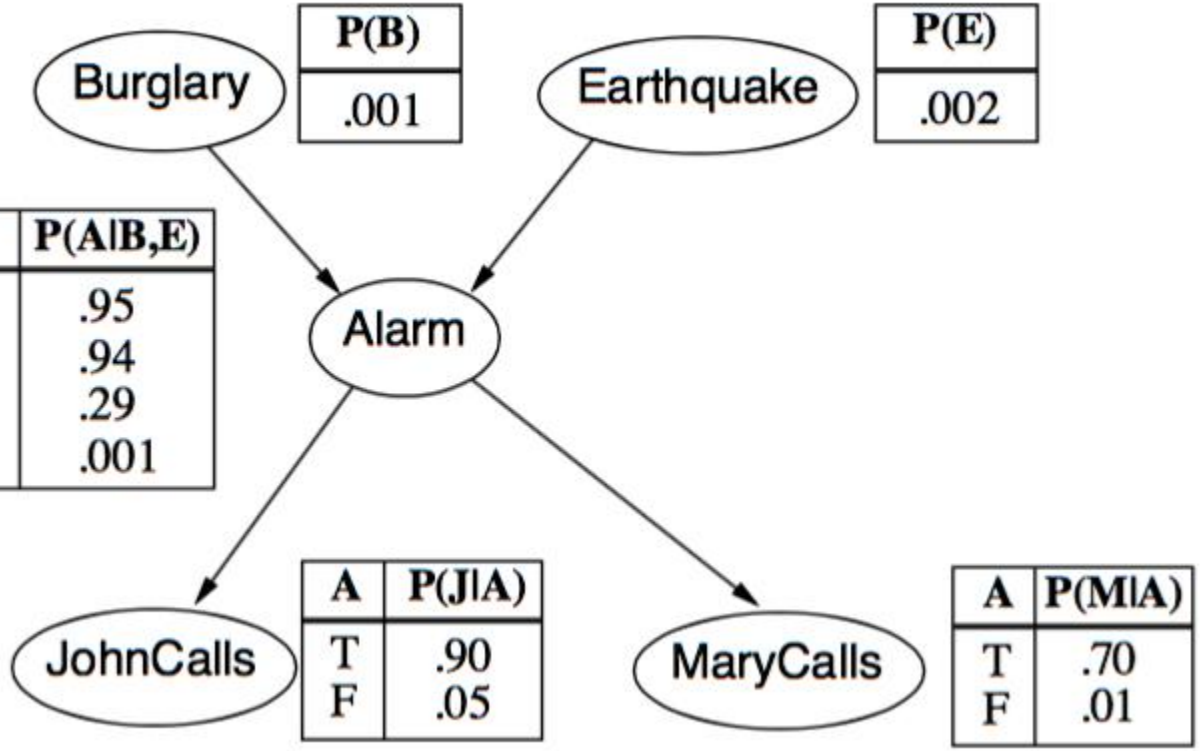
Example: Alarm

Enrich the network with more links: more realistic, less compact



B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

Bayesian Network

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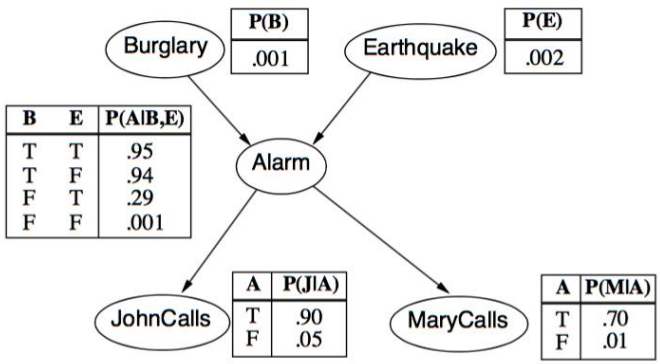

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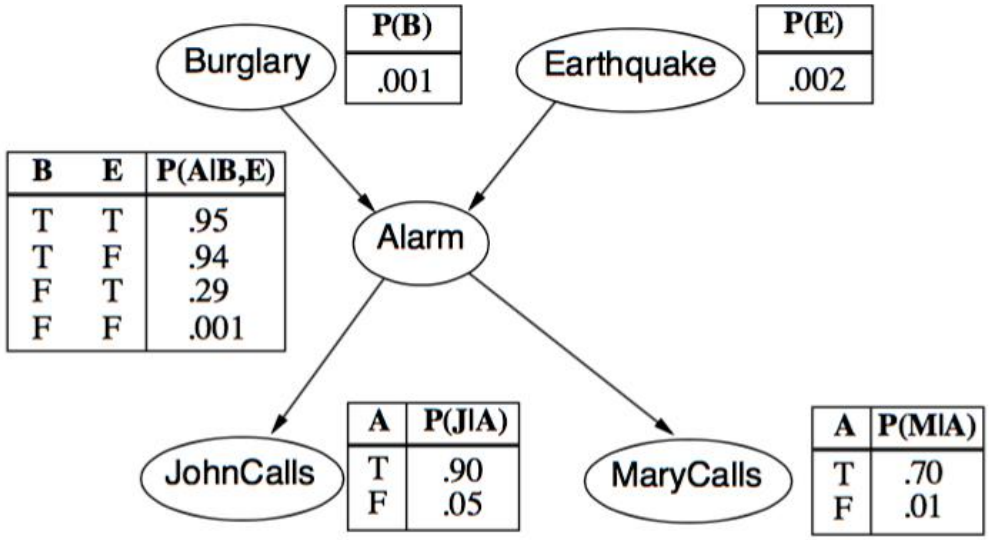
Have to be equivalent!



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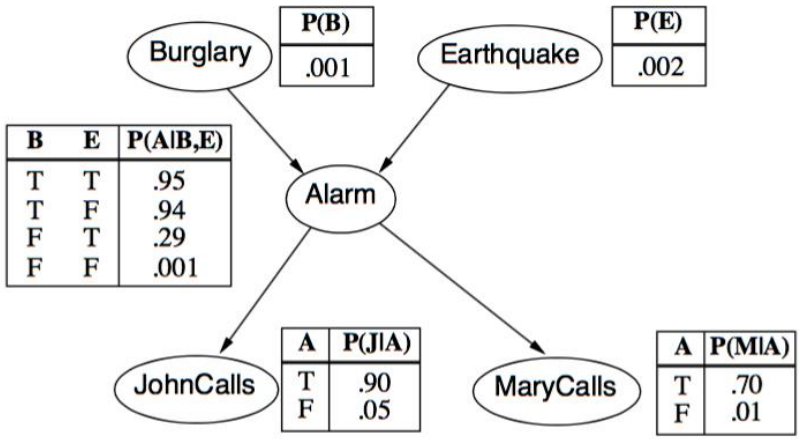

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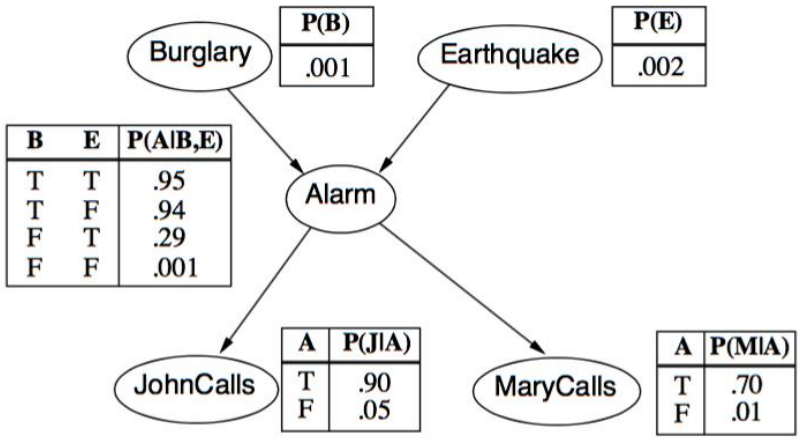
A Bayes' Net = topology (graph) + local conditional probabilities



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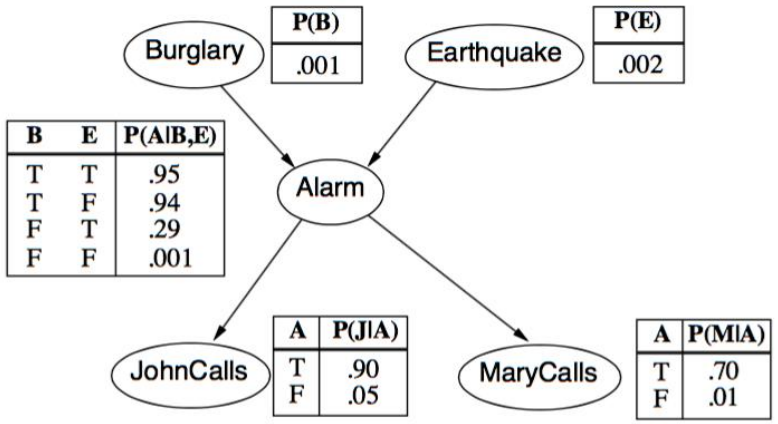

Probabilistic Inference in Bayes' Net

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- Exact Inference in Bayes' Net: (1) Enumeration; (2) Variable Elimination

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This is a joint probability (the value of all random variables are given).

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- Inference with full joint probability distribution table available: Read the joint probability from the table
- Inference in Bayes' Net: compute joint probability through conditional probability table

$$\mathbf{P}(X_1,\ldots,X_n) = \prod_i \mathbf{P}(X_i|Parent(X_i))$$

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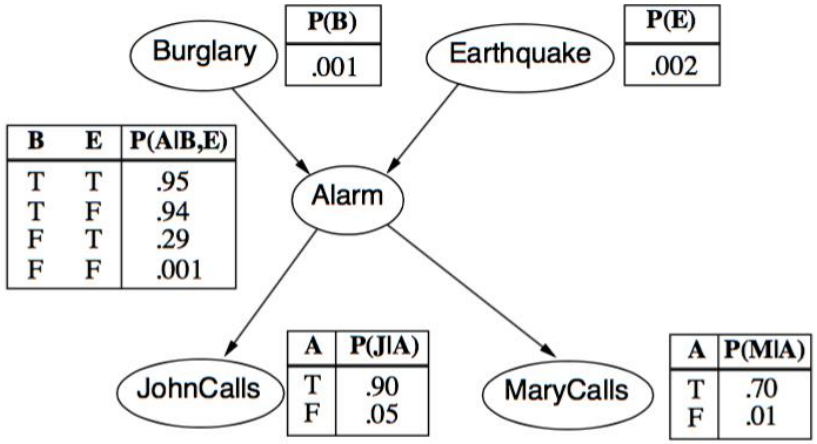

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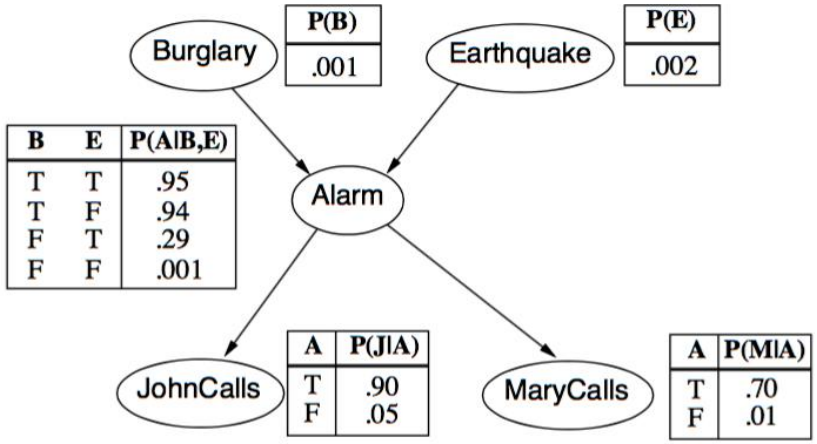

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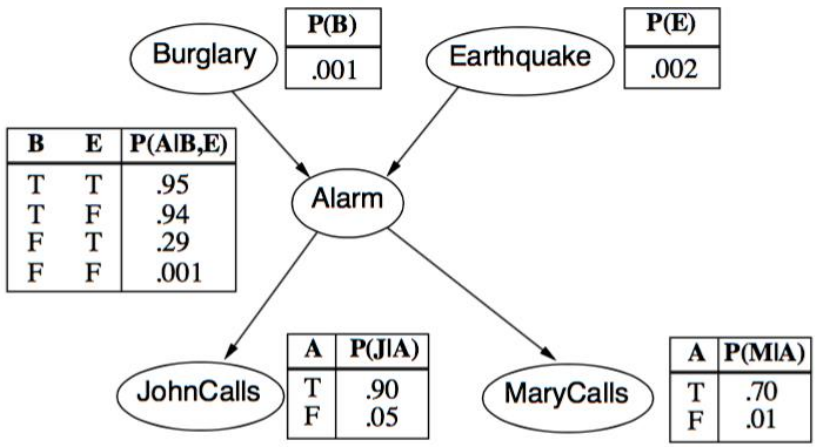
$$\mathbf{X} = \{Burglary\}$$
$$\mathbf{E} = \{JohnCalls, MaryCalls\}$$
$$\mathbf{Y} = \{Alarm, Earthquake\}$$



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- Evaluate through depth-first recursion of the following expression tree

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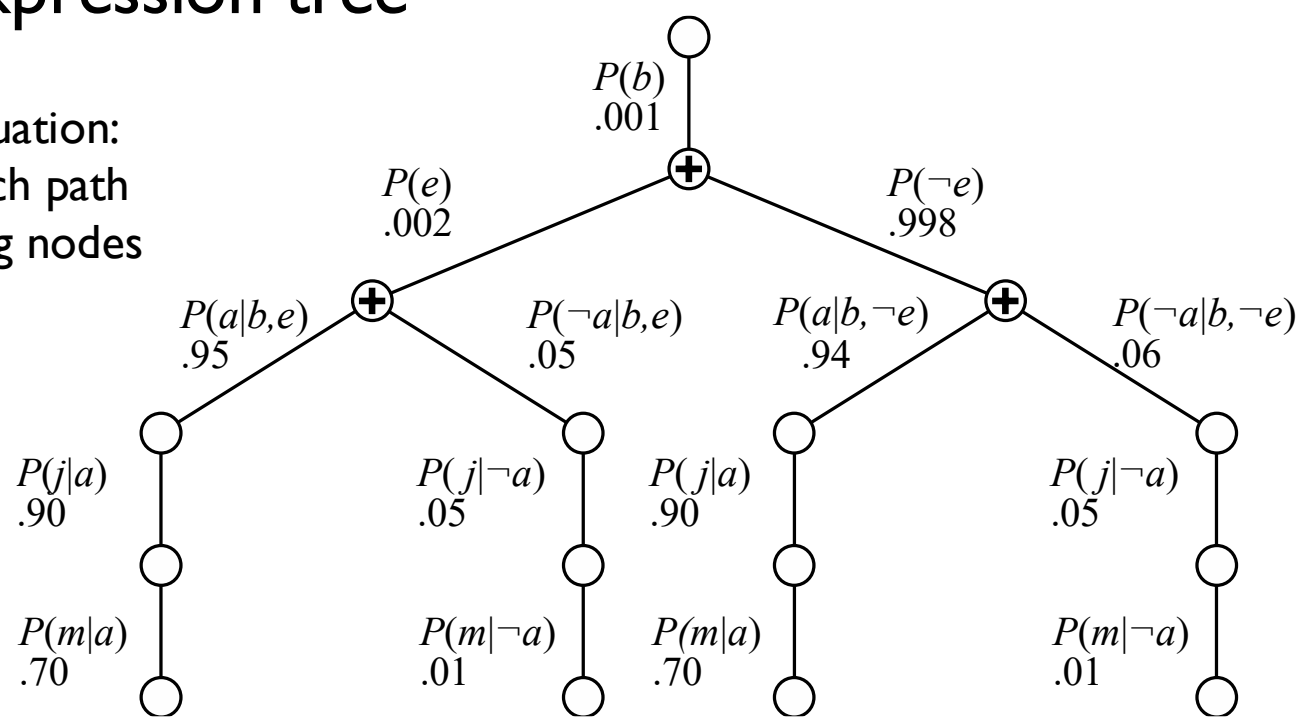
Top-down DF evaluation:
- × Values along each path
- + at the branching nodes



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- Normalize

$$P(b|j,m) = \frac{0.00059224}{0.00059224 + 0.0014919} \approx 0.284$$

$$P(\neg b|j,m) \approx 0.716$$

Exact Inference in Bayes' Net: Enumeration

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- Enumeration: Evaluate

$$\mathbf{P}(\mathbf{X}|\mathbf{e}) = \alpha\sum_{\mathbf{y}\in\mathbf{Y}}\mathbf{P}(\mathbf{X},\mathbf{e},\mathbf{y})$$

This means we need to compute for all the $\mathbf{x} \in domain(X_1) \times domain(X_2) \times \cdots$

We know $\sum_{\mathbf{x}\in\mathbf{X}}\mathbf{P}(\mathbf{X}|\mathbf{e}) = 1$. So we can keep α in the calculation without giving it any actual number, and normalize in the end

using depth-first recursion, then normalize

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Outline

- Bayesian Networks
- Independence in Bayes' Net
- Construct a Bayes' Net
- Exact Inference in Bayes' Net
- Applications of Bayes' Net

Bayes' Net as a Model of Real World

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- Bayes' Net represents knowledge in an uncertain domain
- View it as a way to model the real world based on domain knowledge
- Is your model (Bayes' Net) for a real-world problem correct? Not necessarily.

Bayes' Net as a Model of Real World

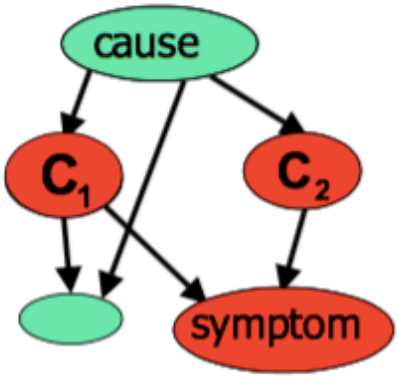
- "All models are wrong"
  - A common aphorism in statistics
  - Generally attributed to the statistician George Box

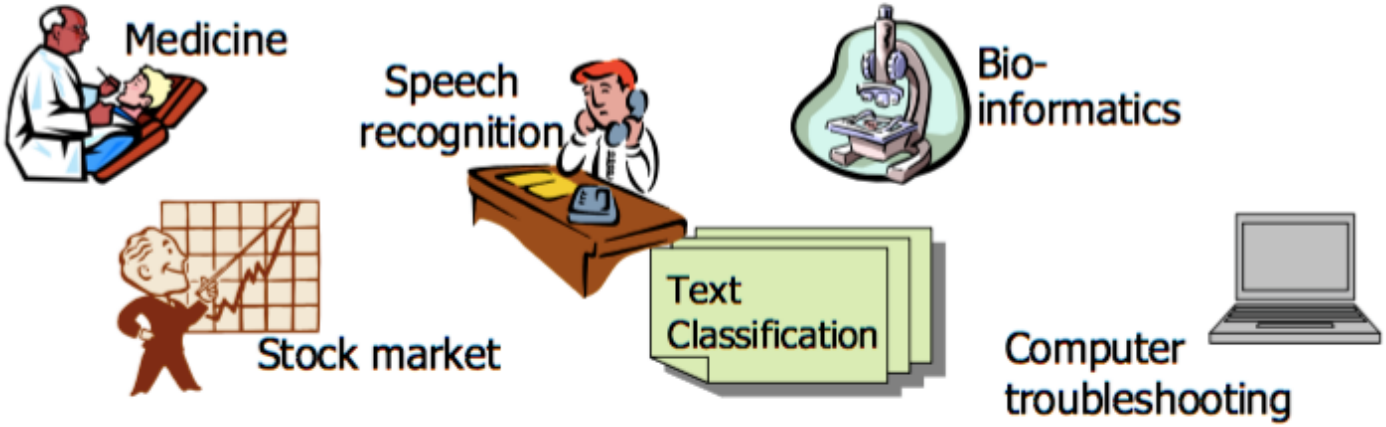
"Essentially, all models are wrong, but some are useful".

https://en.wikipedia.org/wiki/All_models_are_wrong

Use of Bayes' Net

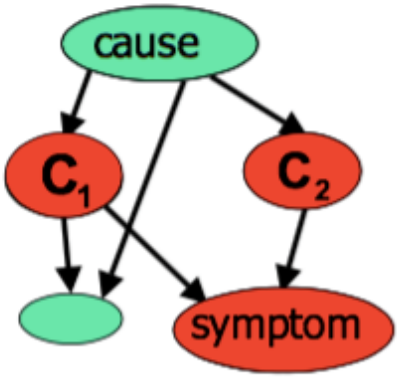
- Diagnosis: $P(\text{cause} \mid \text{symptom})$?

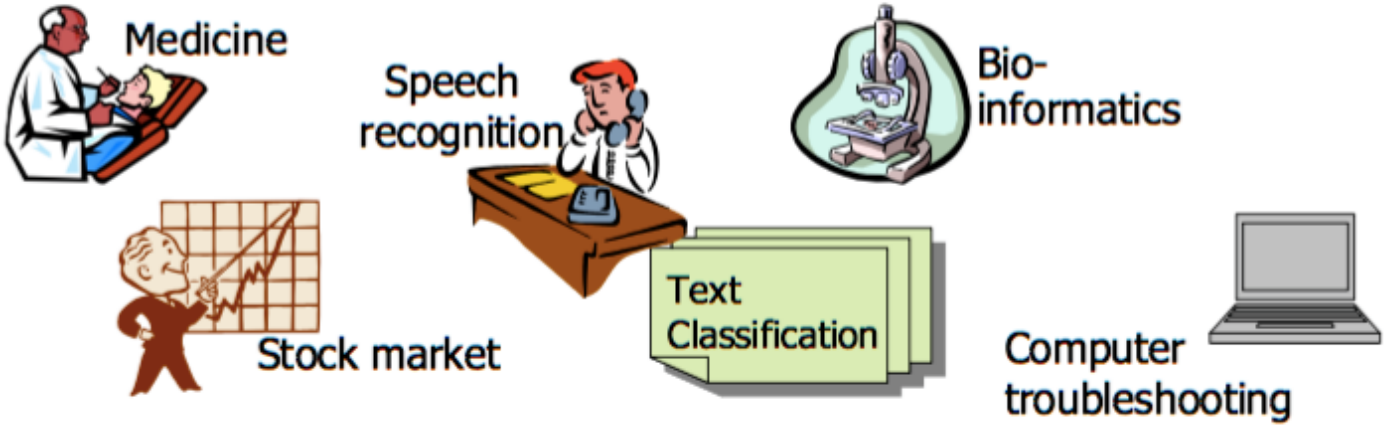




Use of Bayes' Net

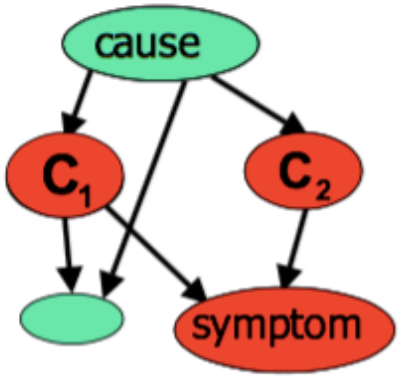
- Diagnosis: $P(\text{cause} \mid \text{symptom})$?
- Prediction: $P(\text{symptom} \mid \text{cause})$?

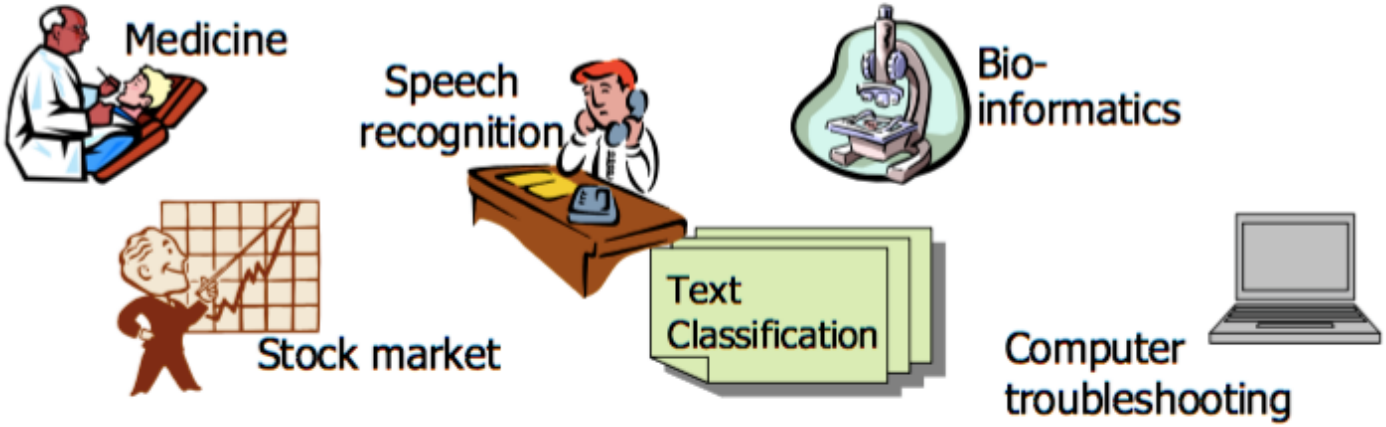




Use of Bayes' Net

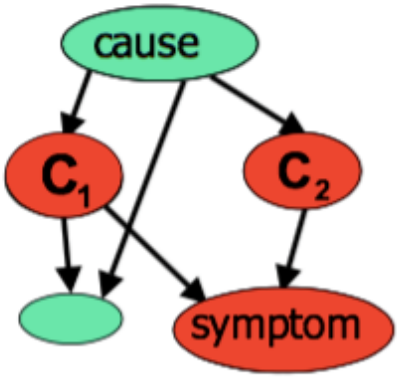
- Diagnosis: $P(\text{cause} \mid \text{symptom})$?
- Prediction: $P(\text{symptom} \mid \text{cause})$?
- Classification: $\max_{class} P(\text{class} \mid \text{data})$

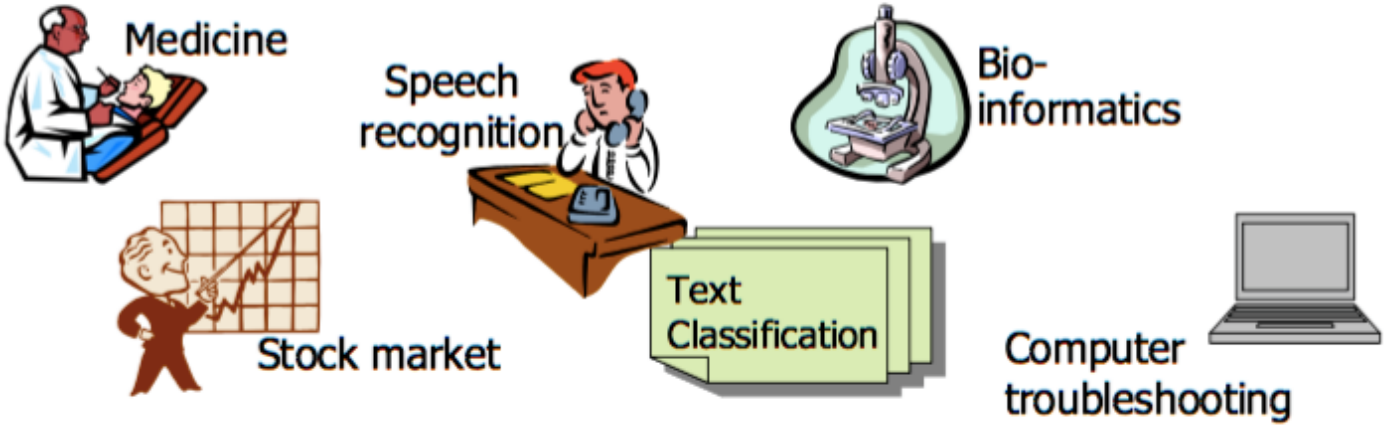




Use of Bayes' Net

- Diagnosis: $P(\text{cause} \mid \text{symptom})$?
- Prediction: $P(\text{symptom} \mid \text{cause})$?
- Classification: $\max_{class} P(\text{class} \mid \text{data})$
- Decision-making (given a cost function)

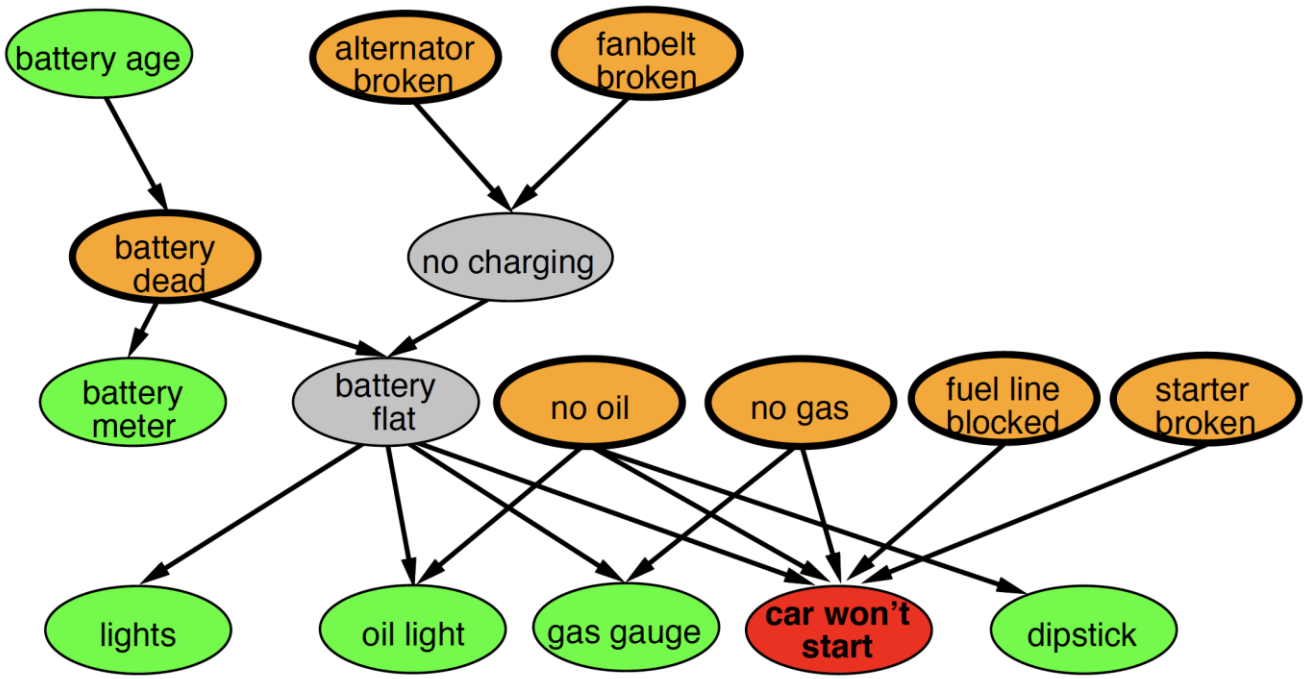




Use of Bayes' Net

Example: Car diagnosis

Initial evidence: car won't start
Testable variables (green), "broken, so fix it" variables (orange)
Hidden variables (gray) ensure sparse structure, reduce parameters



Russel and Novig

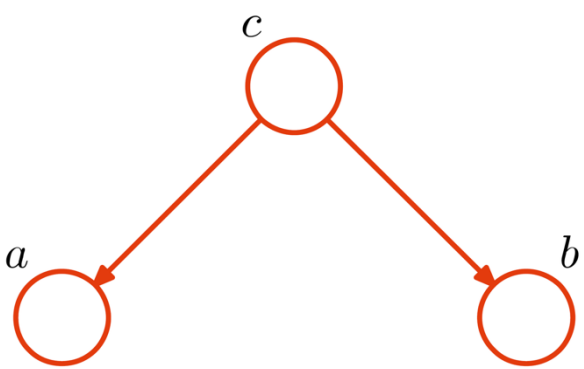
Summary

- Bayes' Net
  - Graphical model
  - Decompose full joint probability distributions into interpretable, simple, local distributions
- Independence in Bayes' Net
  - Local semantics
  - Markov Blanket
- Construct a Bayes Net
- Exact Inference in Bayes' Net
- Applications of Bayes' Net

Backup Slides

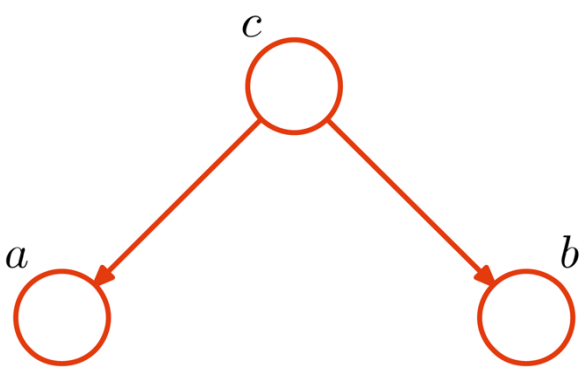
Conditional Independence

- Example graph (1)



Conditional Independence

- Example graph (1)



$$\mathbf{P}(a,b,c) = \mathbf{P}(c)\mathbf{P}(a|c)\mathbf{P}(b|c)$$

$$\mathbf{P}(a,b|c) = \frac{\mathbf{P}(a,b,c)}{\mathbf{P}(c)} = \mathbf{P}(a|c)\mathbf{P}(b|c)$$

So $a \perp b|c$.

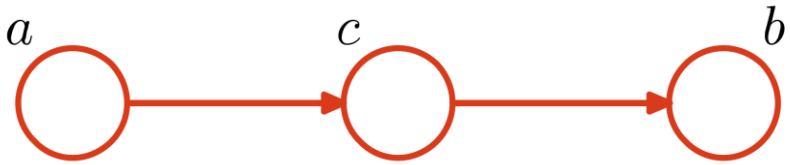
$$\mathbf{P}(a,b) = \sum_c \mathbf{P}(a,b,c) = \sum_c \mathbf{P}(c)\mathbf{P}(a|c)\mathbf{P}(b|c)$$

Cannot be rewritten in the form of $\mathbf{P}(a,b) = \boldsymbol{\theta}_1(a)\boldsymbol{\theta}_2(b)$

So, $a \perp b|\emptyset$ is not always valid

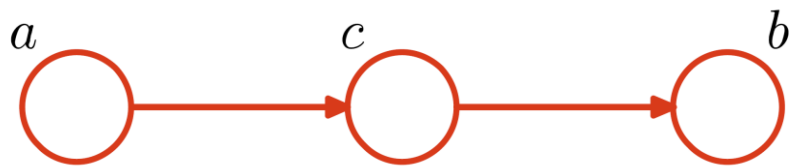
Conditional Independence

- Example graph (2)



Conditional Independence

- Example graph (2)



Bayes' Rule

$$\mathbf{P}(a,b,c) = \mathbf{P}(a)\mathbf{P}(c|a)\mathbf{P}(b|c) = \frac{\mathbf{P}(a)\mathbf{P}(a|c)\mathbf{P}(c)\mathbf{P}(b|c)}{\mathbf{P}(a)} = \mathbf{P}(a|c)\mathbf{P}(c)\mathbf{P}(b|c)$$

$$\mathbf{P}(a,b|c) = \frac{\mathbf{P}(a,b,c)}{\mathbf{P}(c)} = \mathbf{P}(a|c)\mathbf{P}(b|c)$$

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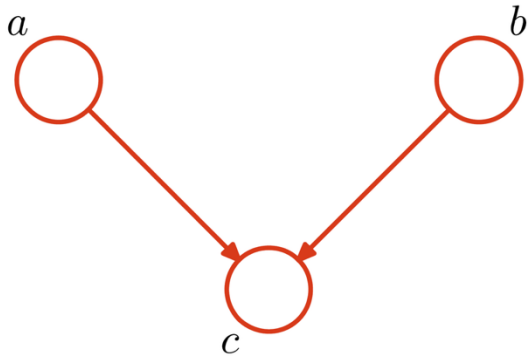
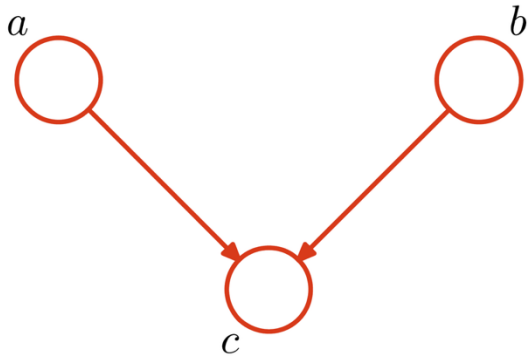
Conditional Independence

- Example graph (3)



Conditional Independence

- Example graph (3)



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$$\mathbf{P}(a,b|c) = \frac{\mathbf{P}(a,b,c)}{\mathbf{P}(c)} = \frac{\mathbf{P}(a)\mathbf{P}(b)\mathbf{P}(a,b|c)}{\mathbf{P}(a,b)}$$

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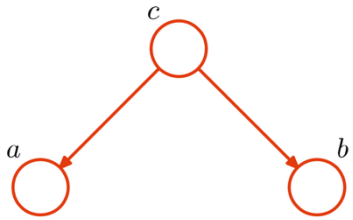
$$\mathbf{P}(a,b) = \sum_c \mathbf{P}(a,b,c) = \sum_c \mathbf{P}(a)\mathbf{P}(b)\mathbf{P}(c|a,b) = \mathbf{P}(a)\mathbf{P}(b)\overbrace{\boxed{\sum_c \mathbf{P}(c|a,b)}}^{=1} = \mathbf{P}(a)\mathbf{P}(b)$$

So, $a \perp b|\emptyset$.

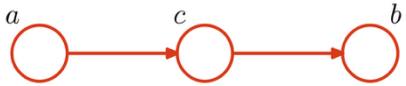
D-Separation for Conditional Independence

- $A \perp B | C$ is valid in general if and only if all the paths from any node in A to any node in B are blocked
- A path is blocked if and only if it includes a node such that either one of the following statements are true
  - The rows on the path meet head-to-tail or tail-to-tail at the node, and the node is in the set C
  - The rows on the path meet head-to-head at the node, and neither the node nor any of its descendants, is in the set C

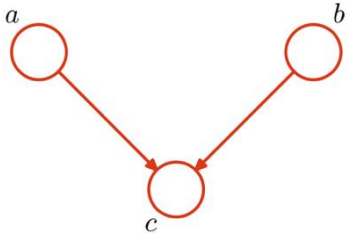
Tail-to-tail at c



Head-to-tail at c



Head-to-head at c



D-Separation for Conditional Independence

- How to tell if $A \perp B | C$ based on D-Separation?
  - Set $independence \leftarrow true$
  - For each path from $a \in A$ to $b \in B$ (Loop 1)
    - Set $blocked \leftarrow false$
    - For each node c on the path from a to b (Loop 2)
      - If head-to-tail or tail-to-tail
        - If $c \in C$
          - $blocked \leftarrow true$
          - Break (Loop 2)
      - If head-to-head
        - $testFlag \leftarrow false$
        - For each node $d \in \{c \cup$ descendant of $c\}$ (Loop 3)
          - If $d \in C$
            - $testFlag \leftarrow true$
            - Break (Loop 3)
        - If $testFlag = false$
          - $blocked \leftarrow true$
          - Break (Loop 2)
    - If $blocked = False$
      - $independence \leftarrow false$
      - Break (Loop 1)
  - Return $independence$